

SE3002 Assignment 02 Report

SE3002 Assignment 02 — Structural Testing of PX4- Autopilot v1.17.0

1. Group details and baseline

Student	Name	Roll number	Primary area
S1	Taimoor Shaukat	24i3015	environment, baseline, coverage, gaps
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Section B, Course SE3002 Software Quality Engineering, Assignment 02.

Baseline: tag v1.17.0, built from commit d6f12ad1c4f70ad3230afd7d86e971421e02fef4 (this is the commit the tag points to via v1.17.0^{commit}, not the annotated tag object hash a5eb12d2ab591251faa009f76b2685b8cc64405d, which is a distinct git object — citing the tag object instead of the commit it resolves to is a documented mistake to avoid, see work/explain/P01.md). Work was carried out on branch sqe-a2, with merge-base with v1.17.0 confirmed equal to the required commit.

2. Local environment and setup evidence

All tooling versions were captured directly, not assumed: Ubuntu 24.04.4 LTS, x86_64, 13th Gen Intel Core i7-1355U (12 cores), 30 GiB RAM, kernel 7.0.0-34-generic. Compiler: GCC/G++ 13.3.0 (Ubuntu 13.3.0-6ubuntu2~24.04.1); cmake 4.4.3; ninja 1.13.2; GNU Make 4.3; Python 3.12.3; lcov/genhtml 2.0-1; gcov 13.3.0; git 2.43.0. No Clang was installed or required — the macOS Coverage-build flag fix documented in the project's reference material does not apply to this Linux/GCC path.

PX4's own setup script (Tools/setup/ubuntu.sh) installs its toolchain via apt-get, which requires sudo; this environment had no non-interactive sudo available (sudo -n true failed). Rather than block on that, every missing tool was installed user-locally instead, with no elevation: cmake/ninja via pip install --user; lcov/genhtml 2.0 built from the upstream lcov source with make install PREFIX=~/.local; the one missing Perl module (DateTime) installed into a user-local local::lib/cpan environment. gcc/g++/gcov and python3 were already present system-wide. ~/.bashrc was updated to put ~/.local/bin on PATH. This is a legitimate, reproducible deviation from the documented sudo-based path, not a shortcut around a requirement; a reader with working sudo access can use the standard Tools/setup/ubuntu.sh script instead (Appendix A documents both).

Build/test commands used throughout: tools/sqe_run_tests.sh build|run|probes|shuffle and tools/sqe_coverage.sh build|baseline|student|final (wrappers around cmake --build build/px4_sitl_test and ctest), never upstream make tests_coverage (which runs make clean, omits branch data, and silently ignores lcov mismatches).

Baseline make tests (upstream suite, unmodified source, before any student test code existed): **100% tests passed, 147 of 147**, total test time 45.73s. No pre-existing failures were observed at baseline.

3. Repository analysis and scope selection record

Four areas (A–D) were selected after re-measuring candidates directly against the live v1.17.0 source (not copied from pre-derived reference material) via grep-based decision/compound-decision counts and an upstream-gtest registration search (119 px4_add_unit_gtest/px4_add_functional_gtest lines, none matching data_validator or failure_detector).

Area A — DataValidator.cpp (src/modules/sensors/data_validator/): tracks one sensor instance's value stream and derives a 0.0-1.0 confidence from timeout, staleness and error-density/count, feeding the Area B voter. 155 lines, 17 decision points, 0 compound decisions. No params, no uORB; all inputs are explicit put()/confidence() arguments. Test level: unit (px4_add_unit_gtest), since no DEFINE_PARAMETERS/uORB subscription/runtime-service construction occurs outside the diagnostic print() method.

Area B — DataValidatorGroup.cpp (MC/DC component): owns a linked list of DataValidators, ranks by confidence × priority, selects the “best” sensor and classifies failover events. 347 lines, 42 decision points, **7 compound decisions** — the densest unit-level candidate. Confirmed call sites: modules/sensors/voted_sensors_update.cpp (IMU accel/gyro voting) and modules/sensors/vehicle_magnetometer/VehicleMagnetometer.cpp (magnetometer voting) — this logic directly decides which sensor instance feeds the state estimator. Test level: unit, same reasoning as A.

Area C — FailureDetector.cpp: detects attitude-envelope exceedance, ESC/telemetry faults, motor under-current, imbalanced-propeller vibration and external-ATS signals, aggregating into Commander's failsafe status bitmask. 354 lines, 46 decision points, **15 compound decisions** — the largest and most logically dense area. 8 params read at construction (FailureDetector.hpp:129-138), 6 uORB subscriptions. Confirmed call site: modules/commander/Commander.cpp:1857. Test level: functional (px4_add_functional_gtest) — params are read once at construction, requiring the set-params-then-construct ordering; gtest_functional_main initializes uORB/params.

Area D — FailureInjector.cpp: reads SYS_FAILURE_EN and injected vehicle_commands to simulate motor failures for in-the-loop testing of Area C. 134 lines, 17 decision points, 4 compound decisions. Test level: functional, same reasoning as C.

Together A-D cover ~990 source lines, ~122 decisions, ~26 compound decisions across two safety chains (sensor redundancy A+B; failure detection C+D), both pure control/business logic with zero existing upstream GTest coverage.

Exclusions (file-spanning, D-004, flagged HUMAN-DECISION for team review): trivial header-only accessors (single-statement returns, no branches); the orphaned src/modules/sensors/data_validator/tests/ directory (confirmed dead — data_validator/CMakeLists.txt never adds it, observation O-01); print() diagnostic-dump methods kept in scope for statement/branch coverage but excluded from MC/DC derivation as non-business-logic output.

Candidates rejected (re-measured against live source, none selected for the final scope): `lib/battery/battery.cpp` (stretch candidate, not needed); `land_detector/*` (heavy module-lifecycle harness, better suited to SITL); `HealthAndArmingChecks/checks/batteryCheck.cpp` (tightly coupled to shared check framework); `commander/failsafe/{failsafe,framework}.cpp` (upstream `failsafe_test.cpp` already exists, too large for the time box); `manual_control/ManualControlSelector.cpp` and `navigator/GeofenceBreachAvoidance` (already upstream-tested); `lib/hysteresis/hysteresis.cpp` (already upstream-tested, used only as a dependency of Area C); `commander/Safety.cpp` and `commander/UserModeIntention.cpp` (too small a decision surface). Re-verification corrected two REF_02 entries during candidate re-measurement: `Safety.cpp` and `UserModeIntention.cpp` were listed there as upstream-tested, but no file-specific gtest registration was found for either — the rejection verdict is unaffected, but the stated reason was corrected for accuracy.

4. Testing approach

Decisions were derived per-file from the live v1.17.0 source into per-area inventories (`work/basis/INVENTORY_DataValidator.md`, `INVENTORY_DataValidatorGroup.md`, `INVENTORY_FailureDetector_Injector.md`) plus three control-flow sketches (`CFG_DataValidator_confidence.md`, `CFG_DataValidatorGroup_get_best.md`, `CFG_FailureDetector_updateMotorStatus.md`) before any test code was written. A setup map (`work/basis/SETUP_MAP.md`) records, per area, exactly how each decision is controlled: A/B take every input as an explicit function argument (`put()`, `confidence()`, `get_best()`); C/D require the ordering `params-set` → `object-constructed` → `uORB-published` → `update()-called`, since `DEFINE_PARAMETERS` is read once at construction and `uORB` has latest-sample (not queued) semantics for single-instance topics.

Oracles follow SPEC_02 §3's allowed sources: return values, getters, published `uORB` messages, flags/bitmasks and counters, derived from code comments/headers/documented library contracts or independently-derived arithmetic — never the production expression re-copied into the test (a circular oracle) and never “does not crash” alone. Several tests are explicitly **characterization tests** (asserting actual behaviour where the specification is silent/ambiguous, labelled characterization (F-nn) in their header comment per SPEC_02 §3): DV07 (all-NaN stream), DV19 (`print()`'s side effect), DV20 (timestamp-0 sentinel) and FD33 (motor-mask-survives-disarm). These are reported as characterizations, not specification-backed oracles, and are cross-referenced to their corresponding findings in §7.

Independence and repeatability were demonstrated per binary with `--gtest_shuffle` and repeat counts of 5 for the two unit binaries and 10 for the three functional binaries; every shuffle run reports zero failures: unit-SqeDataValidator 20/20 PASSED ×5, unit-SqeDataValidatorGroup 34/34 PASSED ×5, functional-SqeFailureDetector 36/36 PASSED ×10, functional-SqeFailureDetectorImu 8/8 PASSED ×10, functional-SqeFailureInjector 13/13 PASSED ×10 (the FailureDetector/Imu counts reflect SQE-FD-36/37 and SQE-FDI-08, added in the P10 coverage-iteration phase after the original P09 shuffle evidence was captured; the logs cited here were re-run against the final test set, not left stale). Each test also passes individually under `--gtest_filter`.

The full 1:1 test-to-decision mapping (116 inventory rows: 20 SQE-DV, 12 SQE-DVG, 22 SQE-DVG-MC, 36 SQE-FD, 8 SQE-FDI, 13 SQE-FI, 5 SQE-PRB probes) is not duplicated here — see `work/inventory/test_inventory.csv` and the “Test Inventory” sheet of the submitted workbook (`deliverables/24i3015_24i3166_24i3158_B.xlsx`).

5. MC/DC component selection and interpretation

DataValidatorGroup was chosen for the assignment’s one required MC/DC component because, among the four scoped areas, it combines the highest compound-decision density at the unit level (5 genuinely compound Boolean decisions, 2-7 atomic conditions each, pure functions of explicit arguments/internal state with no params/uORB/hidden I/O) with the clearest safety argument: its `get_best()` decides which physical sensor feeds the EKF state estimator, and `failover_index()/failover_state()` report when and why a failover occurred. FailureDetector has more compound decisions in absolute count but is covered to decision/branch level only, per the assignment’s “one justified component” requirement (D-003).

The 5 decisions: DVG-D13 (`get_best()` candidate-switch test, L187-190, 7 conditions A-G); DVG-D14 (“did best change”, L203, 3 conditions H/I/J); DVG-D15 (“real failsafe vs priority preference”, L207-208, 3 conditions K/L/M); DVG-D32/DVG-D34 (`failover_index()/failover_state()`, L282-283/L301-302, identical 3-condition pattern P/Q/R). 29 target-evaluation rows across 22 distinct test IDs implement these five decisions with **unique-cause independence pairs under short-circuit relaxation**: in every pair the target condition flips, the decision outcome flips, and every other condition that is evaluated in both tests holds the same value — but, because C++ `&&/||` short-circuit, a condition may be evaluated in one test of a pair and not evaluated at all in the other (recorded as NE(x) in the matrix). Example: in the A pair SQE-DVG-MC-07/08, C is not evaluated in MC07 (A&&B already

true) but is evaluated True in MC08. This is the standard short-circuit form of unique-cause MC/DC (no condition that is evaluated in both tests changes), not strict unique-cause with all other conditions identical, and not masking MC/DC (which would allow evaluated conditions to change).

Three worked examples. The **E pair** (DVG-D13, `fabsf(confidence-max_confidence)<0.01f`): SQE-DVG-MC-01 seeds sensor 0 at confidence 0.95/priority 50, then introduces sensor 1 at confidence 0.95/priority 75 ($|c-m|=0<0.01f$, $E=True$); its independence pair SQE-DVG-MC-03 changes only the candidate's confidence to 0.93 ($|0.93-0.95|=0.02f\geq0.01f$, $E=False$), holding every other condition's evaluated value fixed, and the decision outcome flips (switch vs no switch).

The **K pair** (DVG-D15, `pre_check_prio != -1`): when $K=False$ (no prior "best" was ever seeded), the outer if (`best != nullptr`) guard at L213 is structurally forced reachable-but-always-True by a separate dominance proof (see §6 G-03), and `_curr_best<0` at L219 makes the `true_failsafe` value irrelevant to any getter — so the $K=False$ row (SQE-DVG-MC-21) cannot be distinguished from its True counterpart (SQE-DVG-MC-01) by any API return value. This row is classified **O3** (structural evidence only): line 210 (`true_failsafe = false;`) shows 0 hits for MC21 versus 1 hit for MC01 in their respective per-test `.info` captures, which is the only available proof the test reached its intended branch.

The **P pair** (DVG-D32/D34, shared by `failover_index()/failover_state()`): SQE-DVG-MC-13 and SQE-DVG-MC-25 form the P pair; SQE-DVG-MC-25 specifically constructs the "put with timestamp 0" scenario (`DataValidator.cpp`'s timestamp-0 sentinel, F-12) to drive the validator's `used()` state to False, which flips $P = next \rightarrow used()$ while Q and R's evaluated values are held fixed.

Three condition combinations in DVG-D13 and one in DVG-D14 were determined **infeasible (with proof)**, not simply omitted: $B=T \wedge G=F$ and $F=T \wedge D=F$ are excluded because $B \Rightarrow G$ and $F \Rightarrow D$ are real implications of the underlying float/priority comparisons; $A=T \wedge B=T \wedge C=F$ is excluded because $(A \wedge B) \Rightarrow C$; and DVG-D14's $H=F \wedge I=F \wedge J=F$ is excluded by an algebraic trace showing $J=F$ forces $I=T$. A related float-precision characteristic (F-06: the 1% tie-break's $1.0f - d/100.0f$ is not exact at integer d , confirmed by direct float32 arithmetic, e.g. $d=10 \rightarrow 0.89999998f$) is documented but deliberately not exercised by any of the 12 chosen D13 vectors, each kept ≥ 0.01 – 0.02 away from the boundary so the suite's PASS/FAIL results are stable across runs.

The matrix was independently re-derived by hand and cross-checked with a standalone Python mirror of the exact C++ operator/short-circuit order; `tools/sqe_mcdc_check.py` reports ALL

DECISIONS COMPLETE for all 5 decisions, confirming every condition has both a True-outcome and False-outcome evaluation with a traced unique-cause independence pair.

6. Coverage analysis (baseline vs final, raw vs feasible)

Baseline (upstream tests only, before any student test code): DataValidator.cpp 43/58 lines (74.1%), 21/30 branches (70.0%); DataValidatorGroup.cpp 101/155 lines (65.2%), 54/116 branches (46.6%); FailureDetector.cpp and FailureInjector.cpp both 0/153 and 0/62 lines (0.0%/0.0%) and 0 branches, since no upstream test starts Commander.

Final (full student suite, 5 binaries): DataValidator.cpp 58/58 lines (100.0%), 30/30 branches (100.0%); DataValidatorGroup.cpp 154/155 lines (99.4%), 110/116 branches (94.8%); FailureDetector.cpp 153/153 lines (100.0%), 189/254 branches (74.4%); FailureInjector.cpp 62/62 lines (100.0%), 47/57 branches (82.5%). Total across the scope: 427/428 lines (99.8%), 376/457 branches (82.3%). A separate student-only capture (excluding the upstream suite entirely) produced identical numbers, confirming the student suite alone drives all of the scope's measured coverage, not an interaction with pre-existing upstream tests.

This is gcov/lcov's **raw** tool number. Per the course's stated gcov semantics, "branch coverage" here is condition-level coverage of evaluated operands for short-circuit &&|| expressions (one branch pair per short-circuit operand), not full MC/DC — the MC/DC matrix in §5 is the independent, stronger claim for DataValidatorGroup specifically.

Raw vs exception-filtered branch coverage. Because the Coverage build has C++ exceptions enabled, GCC also instruments an exception-unwind edge after almost every function call (lcov block IDs e0; gap G-05 below). These are not decisions in the source. Counting only the source-level decision/condition branches (every BRDA entry whose block ID does not start with e, computed directly from the same final/scope.info):

File	Raw branches	Exception edges	Source-level branches
DataValidator.cpp	30/30 (100.0%)	0	30/30 (100.0%)
DataValidatorGroup.cpp	110/116 (94.8%)	0	110/116 (94.8%)
FailureDetector.cpp	189/254 (74.4%)	64 (0 hit)	189/190 (99.5%)

File	Raw branches	Exception edges	Source-level branches
FailureInjector.cpp	47/57 (82.5%)	9 (0 hit)	47/48 (97.9%)
Scope total	376/457 (82.3%)	73 (0 hit)	376/384 (97.9%)

All 73 exception edges are uncovered (no PX4 call in the scope throws), and the 8 remaining uncovered source-level branches are exactly the items of G-01, G-02, G-03, G-04, G-06 and G-07 below. The raw 82.3% remains the tool's headline number; 97.9% is the figure that measures decision coverage of the source. Both numbers were independently reproduced on a second machine (fresh Ubuntu 24.04 WSL2 install, clean v1.17.0 clone, this patch applied, plain `make/ctest/lcov` commands rather than the project wrapper scripts): the per-line and per-branch hit sets were identical to `evidence/coverage/final/scope.info` for all four files.

Coverage was reached in three iterations after the IT-0 baseline. **IT-1** (first full capture with all 5 binaries) investigated every uncovered item via raw `gcov -b -c` re-derivation (not just `lcov`'s BRDA summary) and classified most as tool-artefact or infeasible, while identifying 3 genuine missing-test gaps (FD-D25, FD-D38, FD-D41). **IT-2** closed those 3 gaps with 3 new tests (SQE-FDI-08, SQE-FD-36, SQE-FD-37), raising `FailureDetector.cpp` branches from 185/254 (72.8%) to 189/254 (74.4%) with no other file's numbers changing. **IT-3** was a confirmatory re-run (no test changes) that reproduced IT-2's numbers byte-for-byte, ruling out leftover `.gcda` state as an artefact of the debugging session.

Raw vs feasible. `work/GAPS.md` documents 7 clusters (G-01...G-07) covering every item left uncovered after IT-3, each with location, class (environment-limited / infeasible-with-proof / tool-artefact) and a written proof or reproduction path — none excluded merely to raise a percentage, and no `LCOV_EXCL_*` marker was added to production code. Two items are **infeasible with proof**: G-03 (`DataValidatorGroup.cpp:213`'s `best != nullptr` `False` side — the same code-flow dominance argument used for DVG-D15's `K=False` row in §5 shows best is unconditionally non-null whenever this guard is reached) and G-06 (`FailureInjector.cpp:43-44`'s `param_get()` failure branch — `SYS_FAILURE_EN` is a compile-time-registered parameter, so `param_find()` on this exact literal name cannot fail in a successfully-linked binary). G-07 is infeasible specifically in a single-threaded test environment (`FailureDetector.cpp:194,222`'s `copy()` `False` immediately after `.updated()` `True` — traced through the real `uORB DeviceNode/Subscription` implementation to show `.updated()⇒_data≠nullptr` in the absence of concurrent writers). One item is **environment-limited**: G-01 (`DataValidatorGroup.cpp:88`'s `-fcheck-new` null-check — real on a non-throwing NuttX allocator, dead code on this

build's throwing operator new ABI; not reached without a dedicated allocator-fault build, not attempted this session). Three are **tool-artefact**: G-02 (delete's compiler-emitted null-guard on a loop-invariant-guaranteed-non-null pointer), G-04 (gcov's documented line-attribution collapse of 6 ternary conditions in print()'s multi-argument PX4_INFO_RAW call onto one source line — independently reproduced with a standalone minimal repro under the same coverage flags) and G-05 (GCC's per-call exception-unwind edges across FailureDetector/FailureInjector, confirmed via gcov -b -c showing taken 0 (throw) on every entry, for library calls that do not throw by PX4 convention). Removing G-05's exception edges gives the 376/384 (97.9%) source-level figure above. If the remaining 8 justified items were also removed, every remaining branch would be covered; that "100% of feasible branches" figure is not used as a headline here, because it depends on accepting each gap proof in work/GAPS.md and G-01 in particular is reachable on a different allocator configuration.

Annotated excerpts (from evidence/coverage/final/html/src/modules/sensors/data_validator/DataValidatorGroup.cpp.gcov.html): at L187-190 (DVG-D13's 7-condition expression), the gcov branch-pair annotations show, e.g., branch 0 (A) taken 454 times and branch 1 (B, only evaluated when A is false) taken 29 times at L187, with every branch pair on these four lines showing a non-zero "taken" count for both outcomes — consistent with the MC/DC matrix's claim of full True/False coverage for all 7 conditions. At L207-213 (DVG-D15's guard and the nested best != nullptr check), L207's K/L/M branches show non-zero counts on both sides, but L213's second branch (best == nullptr) shows tlaUNC/"Branch 1 was not taken" with 0 hits for both sides where L210 (true_failsafe = false;) is executed exactly 2 times total — consistent with G-03's infeasibility proof (the False side of L213 is structurally unreachable) and with the K pair's O3 classification in §5.

7. Findings

All 15 candidate findings from the project's pre-derived register were independently re-investigated this session against the live source and the actual executed test/probe results; **13 had their code-level claim confirmed, 0 rejected, 1 latent and not runtime-verified** (F-13). "Code-level claim confirmed" means the described behaviour really occurs; whether that behaviour is a defect is stated per finding below. Three findings (F-09, F-10, F-11) need a specification-level oracle judgment that the evidence alone cannot settle. The reasoning in work/FINDINGS.md argues F-09 and F-10 are defects and F-11 a specification ambiguity, but that classification was proposed by one team member and has not yet been agreed by the whole team, so F-09 and F-10 are reported below as **candidate defects (pending team review)**, not as

confirmed defects. That zero candidates were rejected is reported as a genuine outcome, not evidence of padding: every REF_07 candidate's underlying code-level claim held up under independent re-reading of the cited file:line this session.

Confirmed defects / observations (title — location — test ID — severity; full reproduction/expected/actual in work/findings/REPORT_BLOCKS.md, not duplicated here): - **F-01** `best != nullptr` guard (`DataValidatorGroup.cpp:213`) is dead code in its False branch — structural proof only, no severity (coverage-measurement artefact, cross-referenced to G-03 in §6/§8). - **F-02** `get_best()` returns a non-null stale-data pointer when `*index == -1` signals total sensor failure, contradicting the header's "array of best values" phrasing — SQE-DVG-11, SQE-DVG-MC-19 confirm; SQE-PRB-03 (disabled probe) fails under the stricter `nullptr` oracle. Latent: both production callers discard the return value and use only `*index`. - **F-03** `DataValidatorGroup(0) + add_new_validator()` dereferences a null `_last` and crashes (confirmed via `EXPECT_DEATH`, SQE-DVG-10) — unreachable from production (all 3 real construction sites use `{1}`). - **F-04** the stale-value counter is shared across all 3 axes rather than per-axis, so a padded scalar sensor (e.g. `airspeed`, `sensors.cpp:285`) reaches "stale" roughly 3× faster than a true 3-axis signal — SQE-DV-05, documented design trade-off. - **F-05** `DataValidator::print()` is not read-only — its internal `confidence()` call can set `ERROR_FLAG_TIMEOUT` as a side effect of printing — SQE-DV-19 (characterization). - **F-06** the 1% confidence tie-break is float-rounding/optimisation-level-sensitive (confirmed by direct float32 arithmetic) — not exercised by the submitted suite's deliberately boundary-avoiding vectors; treated separately as an untested-interaction risk in §8. - **F-07a/F-07b (confirmed at runtime under sanitizers)** an unclamped ESC-count loop and an unguarded shift exponent in `FailureInjector::manipulateEscStatus`. In the normal/Coverage build the probes SQE-PRB-05/06 are inert `SUCCEED()` placeholders and prove nothing; they were therefore run in sanitizer builds of only `FailureInjector.cpp` and the probe file (PX4's own `AddressSanitizer/UndefinedBehaviorSanitizer` build types do not compile on GCC 13 because of the bundled Abseil/fuzztest dependencies). F-07a: ASan reports stack-buffer-overflow, a `READ` at `FailureInjector.cpp:118` (`status.esc[8]`, one past the array) from SQE-PRB-05. F-07b: UBSan reports shift exponent 4294967195 is too large for 32-bit type 'int' at `FailureInjector.cpp:120` from SQE-PRB-06. The 13 active `FailureInjector` tests run clean under both sanitizers. Reachable only through failure injection (`SYS_FAILURE_EN=1`), which is a test feature, so flight impact is limited to injection sessions. - **F-08** the `FAILURE_TYPE_WRONG` log message uses a 0-based motor index while every sibling case uses 1-based — cosmetic, confirmed by direct source read, no dedicated capturing test written (R4 bars manufacturing an assertion to pad the finding). - **F-09 (candidate defect, pending team review)** a stream of only non-finite samples reports `confidence()==1.0/state()==NO_ERROR`, contradicting the class's own

stated purpose of identifying anomalies — SQE-DV-07 (characterization) records the as-shipped behaviour; SQE-PRB-01 encodes the proposed stricter oracle and fails against the as-shipped code. - **F-10 (candidate defect, pending team review)** `_error_density == ERROR_DENSITY_WINDOW` exactly yields `confidence()==0` with no flag set (`>` not `>=` at `DataValidator.cpp:125`) — two outputs of the same object disagree at the same instant. SQE-DV-15 records the as-shipped behaviour; SQE-PRB-02 encodes the proposed stricter oracle and fails; consumer impact traced to `voted_sensors_update.cpp:419`. - **F-11 (specification ambiguity, not a defect)** an equal-priority, confidence-only sensor switch increments `failover_count()` even though the source’s own comment frames the guard as distinguishing “a real failsafe” from “a priority preference” — SQE-DVG-MC-04, SQE-DVG-MC-07 confirm the current, self-consistent behaviour. Unlike F-09/F-10 this has no internal contradiction; resolving it requires knowing what the counter is used for downstream, which this session could not determine from source alone. - **F-12** timestamp 0 is an implicit “no data” sentinel in both `put()` and `confidence()` — SQE-DV-20, SQE-DVG-MC-25 confirm; not reachable in the real system since `hrt_absolute_time()` is never 0 after boot. - **F-15** `FailureDetector`’s `disarm-reset` block clears the under-current mask but never the timed-out mask, so a previously-timed-out motor still reports failed in `getMotorFailures()` after a clean disarm — SQE-FD-33 (characterization).

Two citation corrections to the pre-derived register were made and documented rather than silently fixed: F-02’s originally-cited evidence test (“MC18”) is a stable no-failover control row with no bearing on the finding — the real evidence is MC19; F-12’s originally-cited test (“MC17”) does not exist in the actual suite — the real DVG-level evidence is MC25. Neither correction changes the finding itself.

Rejected candidates: none (0 of 15). **Downgraded:** F-13 (allocation-failure robustness in `DataValidatorGroup`’s constructor/`add_new_validator()`) — the code shape (no null-safe handling reachable on this platform) is confirmed by reading the source, but the claimed runtime crash was not independently reproduced this session (no AF*-series test exists in the submitted suite, and the pre-derived register’s own “pre-verified” claim was made outside the PX4 GTest suite) — reported as latent, not a confirmed runtime defect.

8. Gaps, limitations, residual risk, improvements

The 7 coverage-gap clusters (G-01...G-07, §6) are the primary remaining structural gaps; each has a written proof or reproduction path in `work/GAPS.md` and is not repeated here. In

summary: 1 environment-limited (G-01, NuttX non-throwing-allocator null-check, unreachable under this build's throwing operator `new`), 2 infeasible-with-proof items plus a single-threaded-specific infeasibility (G-03, G-06, G-07), and 3 tool-artefact items (G-02, G-04, G-05, all independently reproduced with standalone minimal repros or raw `gcov -b -c` output, not merely asserted).

Environment limits. All evidence in this report comes from a single compiler/OS pairing — GCC 13.3.0 on Ubuntu 24.04 — with no cross-compiler build and no verification against an actual NuttX target. This matters concretely for G-01: the `-fcheck-new` null-check at `DataValidatorGroup.cpp:88` is dead code on this platform's throwing-operator `new` ABI but is documented as real flight code for targets with a non-throwing (NuttX) allocator configuration — this report cannot claim that code path was exercised on any platform where it is live.

No SITL/HITL. This is a deliberate scope decision (§3, `work/scope/CANDIDATES.md` §4), not an oversight: none of the four scoped files touch drivers, the work-queue scheduler, module start/stop lifecycle or simulator dynamics to reach their decisions, so unit/functional GTest levels reach every decision deterministically without SITL's added nondeterminism and wall-clock cost. The consequence is that this report says nothing about hardware timing, real sensor-driver interaction, or the actual multi-process/multi-thread behaviour of `voted_sensors_update.cpp/Commander.cpp` calling into the scoped files — those call sites are confirmed to exist (§3) but their real runtime interaction with the scoped logic was never executed.

F-06 as an untested-interaction risk. The 1%-confidence tie-break's float-rounding/optimisation-level sensitivity (§5, §7) is a specific, named risk that the coverage and MC/DC results in this report do not bound: the MC/DC matrix's 12 D13 vectors were deliberately kept away from this boundary for test stability, which means the suite provides no evidence about behaviour exactly at that boundary under a different optimisation level (e.g. a `-02/-03` flight build) than the `-00` Coverage build used throughout this session.

Concrete improvements (listed per SPEC_06 §8's guidance; (2) was carried out in the post-audit revision, the rest were not implemented): (1) a dedicated allocation-fault-injection build (SPEC_02 §9's documented GCC/`-fcheck-new` technique, in its own CMake target) to close G-01 and give F-13 runtime evidence; (2) **done:** sanitizer runs of the F-07 probes (§7), now with runtime ASan/UBSan reports; extending that to the whole suite would need PX4's sanitizer build types to be fixed for GCC 13, or a Clang toolchain; (3) a second optimisation-level build (`-02`) to directly measure F-06's adjacent-step divergence instead of relying on hand float32 arithmetic; (4) dependency-level testability improvements such as splitting `print()`'s multi-ternary

PX4_INFO_RAW call (G-04) into separate statements, which — if ever made as a production change outside this assignment's R2 constraint — would let gcov attribute branches correctly per ternary.

9. Final quality judgment

The structural evidence in this report supports specific, bounded claims about the analysed scope — four files, `DataValidator.cpp`, `DataValidatorGroup.cpp`, `FailureDetector.cpp` and `FailureInjector.cpp` — and nothing broader. Measured coverage reached 99.8% of lines and 82.3% of raw branches across the scope; excluding compiler-generated exception edges, 97.9% of source-level branches were covered. `DataValidator.cpp` reached 100.0%/100.0% and `DataValidatorGroup.cpp` 99.4%/94.8%, and every remaining uncovered item is classified with a written proof as infeasible, environment-limited, or a tool-measurement artefact rather than silently dropped. On `DataValidatorGroup`'s five redundancy-selection decisions, full MC/DC was achieved using unique-cause independence pairs under short-circuit relaxation, with every condition's True and False outcomes checked against an exact oracle — return values, state getters, or, for one structurally-unobservable row, documented per-test structural evidence. This combination gives high confidence that every implemented decision outcome in the scope, including each evaluated condition inside `DataValidatorGroup`'s compound expressions, was exercised at least once with a checked oracle.

That confidence does not extend to claims this evidence cannot support. Two findings (F-09, F-10) depend on a specification-level judgment and are reported only as candidate defects pending team review; their probes fail against the proposed stricter oracle, which shows the behaviour but not that it is wrong. F-07a/F-07b, by contrast, are confirmed at runtime by ASan/UBSan. The NuttX non-throwing-allocator path (G-01), hardware timing, and the real multi-threaded/multi-process interaction of the scoped files with their actual callers (`VotedSensorsUpdate`, `Commander`, `VehicleMagnetometer`) were never executed — only the confirmed call sites were read, not exercised end-to-end. F-06's float-rounding sensitivity at the 1% confidence boundary is real but deliberately unexercised by the submitted test vectors.

Condition-level branch coverage plus full MC/DC on the voter's redundancy-selection logic together mean each implemented decision outcome was exercised with a checked oracle — this is not the same claim as “the requirements are complete” or “the code is correct for all inputs,” and this report does not make either of those claims.

What limits any broader conclusion: all testing was at unit/functional GTest isolation from the real sensor/estimator pipeline, on a single compiler and OS (GCC 13.3.0, Ubuntu 24.04), with no SITL or hardware-in-the-loop run. No claim is made, or should be inferred, about PX4 as a whole, about production-build (-02/-03) behaviour, or about any file outside the analysed scope.

10. AI-assistance record

AI (Claude Code) was used materially across all phases: source verification (re-reading production code directly rather than trusting pre-derived reference material before every derivation), build/coverage troubleshooting, test implementation (111 active GTest cases + 5 disabled probes), MC/DC derivation and independent cross-checking, coverage-gap classification, findings investigation, workbook generation, and this report's writing.

Verification practice: every test traces to a decision ID checked by `tools/sqe_trace_check.py` (clean, 0 errors/warnings); every MC/DC pair was independently recomputed against a standalone Python mirror of C++ operator/short-circuit order; every number in this report was re-read from its evidence file this session, not carried from memory. No assertion was weakened to pass a test (R4); the one test-design error found (an early draft's wrong assumption about ESC failure-mask bit layout) was fixed by rewriting the test before any coverage capture used it.

Key assumptions and how they were checked: none taken on faith — every classification in `work/GAPS.md` and `work/FINDINGS.md` required independently reading the cited file:line, tracing further into PX4/uORB internals where needed to construct a real proof. Two explicit, honestly-reported process assumptions: gate approvals (G00-G14) were self-approved autonomously under the user's standing session-start authorization, then later converted to a named human batch sign-off at session end (`work/STATUS.md`'s note states plainly this was a summary-level approval, not sequential per-gate review as intended); the three findings requiring a specification-level oracle decision (F-09, F-10, F-11) were resolved autonomously on one team member's authority, not full team deliberation — flagged for review before the viva, not presented as settled consensus.

Post-audit revision (also AI-assisted, Claude Code): an independent examiner-style audit rebuilt the submission on a second machine and confirmed every coverage number. It then led to these changes: F-07a/F-07b probes run under ASan/UBSan (now runtime-confirmed); exception-filtered branch coverage reported next to the raw figure; MC/DC wording corrected to "unique-cause under short-circuit relaxation"; F-09/F-10 downgraded to candidate defects pending team review; SQE-DVG-12 given real assertions in place of a bare `SUCCEED()`.

Assumption introduced: the AI judged that the sanitizer reports are sufficient runtime evidence for F-07a/F-07b, and that the plain-build PASS of the two probes should not be reported. The team takes responsibility for all targets, expected results, tests and conclusions in this submission.

Appendix A — Reproduction commands

Clone and baseline (work/explain/P01.md):

```
git clone --branch v1.17.0 --recursive <px4-remote> PX4-Autopilot
cd PX4-Autopilot && git rev-parse 'v1.17.0^{commit}' # must
print d6f12ad1c4f70ad3230afd7d86e971421e02fef4
git checkout -b sqe-a2
git submodule status --recursive | grep -c '^-' # must
print 0 (all submodules initialised)
```

Toolchain — standard path (requires working sudo):

```
Tools/setup/ubuntu.sh --no-nuttx --no-sim-tools
```

Toolchain — path actually used this session (no sudo available, D-002):

```
pip install --user cmake ninja
# build lcov/genhtml 2.0 from source: ./configure --
prefix=$HOME/.local && make install PREFIX=$HOME/.local
# install the DateTime Perl module via a user-local local::lib +
cpan, not system-wide
echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc
```

Baseline build and test (P02):

```
make tests TESTFILTER=__no_tests__ # build only
make tests # build + run (config
px4_sitl_test, ctest)
```

Coverage pipeline (P03-P10), via the project's own wrapper (never upstream make tests_coverage):

```
tools/sqe_coverage.sh build
tools/sqe_coverage.sh baseline # pre-student-test
capture
make tests PX4_CMAKE_BUILD_TYPE=Coverage TESTFILTER=__no_tests__
# switch build dir to Coverage
tools/sqe_run_tests.sh build # build all Sqe* test
binaries
tools/sqe_run_tests.sh run # ctest -R Sqe
tools/sqe_run_tests.sh shuffle # --gtest_shuffle --
gtest_repeat=5|10 per binary
tools/sqe_run_tests.sh probes # --
gtest_also_run_disabled_tests
```

```
tools/sqe_coverage.sh student <tag>          # student-suite-only
capture
tools/sqe_coverage.sh final <tag>            # full final capture
(HTML + .info)
```

Sanitizer runs of the F-07 probes (post-audit; run from the Coverage build directory after the build above — each script compiles only FailureInjector.cpp (+ the probe file for ASan) with the sanitizer into /tmp, links a separate test binary, and leaves the Coverage build untouched):

```
PX4_BUILD=PX4-Autopilot/build/px4_sitl_test bash tools/
sqe_asan_probes.sh # PRB-05 -> ASan overflow, L118
PX4_BUILD=PX4-Autopilot/build/px4_sitl_test bash tools/
sqe_ubsan_probes.sh # PRB-06 -> UBSan shift, L120
```

Equivalent plain commands without the wrapper scripts (used for the independent re-run, evidence/repro_independent/):

```
make tests PX4_CMAKE_BUILD_TYPE=Coverage
# baseline: clean v1.17.0, 147 upstream tests
LC="--rc branch_coverage=1 --ignore-errors
mismatch,inconsistent,unused,negative,empty,source,gcov"
lcov $LC --capture --directory build/px4_sitl_test -o all.info
lcov $LC --extract all.info '*/data_validator/DataValidator.cpp'
'*/data_validator/DataValidatorGroup.cpp' \
'*/failure_detector/FailureDetector.cpp' '*/failure_detector/
FailureInjector.cpp' -o scope.info
git apply 24i3015_24i3166_24i3158_B.patch
make tests PX4_CMAKE_BUILD_TYPE=Coverage TESTFILTER=__no_tests__
# build student tests
lcov --zerocounters --directory build/px4_sitl_test
(cd build/px4_sitl_test && ctest -R Sqe)
# student-only run, then capture as above
```

MC/DC and traceability checks (P06, P12):

```
python3 tools/sqe_mcdc_check.py work/mcdc/mcdc_matrix.csv
python3 tools/sqe_trace_check.py --px4 PX4-Autopilot --inventory
work/inventory/test_inventory.csv \
--mcdc work/mcdc/mcdc_matrix.csv
```

Workbook and report:

```
python3 tools/sqe_workbook.py          # generates deliverables/
<BASE>.xlsx
python3 tools/sqe_word_count.py deliverables/report/REPORT.md
```

Appendix B — Evidence index

Path	What it is
evidence/env/environment.md	tool/OS/compiler versions
evidence/baseline/summary.txt	baseline make tests result (147/147)
work/scope/SCOPE_RECORD.md, work/scope/CANDIDATES.md	scope selection and rejected candidates
work/basis/SETUP_MAP.md, work/ basis/CFG_*.md, work/basis/ INVENTORY_*.md	test-basis derivation
work/mcdc/MCDC_ANALYSIS.md, work/mcdc/mcdc_matrix.csv, work/ mcdc/mcdc_check_report.md	MC/DC derivation and checker confirmation
evidence/coverage/baseline/ per_file.md	baseline coverage
evidence/coverage/final/ per_file.md, evidence/coverage/ final/html/	final coverage (table + HTML)
evidence/coverage/ compare_baseline_final.md	baseline vs final comparison
work/coverage_iterations.md	IT-0..IT-3 iteration history
work/GAPS.md	7 gap clusters with proofs
evidence/coverage/pertest/ MC21_03_EVIDENCE.md, MC21_evidence.info, MC01_evidence.info	O3 structural-coverage evidence for the K pair
work/findings/REPORT_BLOCKS.md, work/FINDINGS.md	findings register and report- ready text
evidence/tests/xml/*.xml	per-binary GTest XML results
evidence/tests/logs/ *_shuffle.log	shuffle/repeat stability logs
evidence/tests/probes/*	disabled-probe execution evidence
evidence/tests/sanitizer/*.log	ASan/UBSan runs of SQE- PRB-05/06 and of the 13 active FailureInjector tests
evidence/coverage/post_audit/	coverage re-capture after the SQE-DVG-12 change (unchanged: 427/428, 376/457)
evidence/repro_independent/	independent second-machine reproduction (baseline/ student/final .info, shuffle and probe logs)

Path	What it is
work/inventory/ test_inventory.csv	full test-to-decision mapping (116 rows)
deliverables/ 24i3015_24i3166_24i3158_B.xlsx	testing workbook (2 sheets)
work/ai_assistance_log.md	full AI-assistance log
work/STATUS.md	phase/gate table, HUMAN- DECISION items
work/DECISIONS.md	ADR-lite decision log (D-002, D-004, D-008, D-009, D-010 cited above)